

Structural Music Understanding via Graph Neural Networks and Multimodal Cross-Attention Fusion

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Abstract

Music audio exhibits complex multi-scale relational structures spanning temporal transitions, harmonic progressions, and recurring thematic motifs. Conventional sequence models (e.g., 2D CNNs operating on time-frequency spectrograms) treat audio as static planar images, struggling to capture long-range relational topologies such as repeating verse-chorus patterns. In this work, we present a structural graph-based framework for music context understanding. We model audio tracks as topological segment graphs where nodes represent temporal audio windows enriched with pitch-class chroma and timbral MFCC features, and edges encode sequential continuity and acoustic similarity shortcuts. We develop an inductive **GraphSAGE** structural encoder, benchmark it against a 2D CNN mel-spectrogram baseline, and introduce a **Scaled Dot-Product Cross-Attention Fusion** mechanism that dynamically integrates graph structural representations with contextual semantic text embeddings. Finally, we formulate a dual-encoder **InfoNCE contrastive learning** objective for cross-modal zero-shot retrieval. Evaluated on the **FMA-Small** and **MagnaTagATune** benchmarks, our structural GNN outperforms the 2D CNN baseline by 11 percentage points in Macro-F1 (0.52 vs. 0.41), while the multimodal cross-attention fusion achieves state-of-the-art context tagging (**Macro-F1 of 0.61, mean AUC-PR of 0.55**) and robust bidirectional audio-text retrieval (R@5 of 0.38).

1 Introduction

Deep learning architectures in Music Information Retrieval (MIR) have conventionally relied on 2D Convolutional Neural Networks (CNNs) applied directly to log-mel spectrogram representations. While convolutional kernels effectively capture local spectro-temporal features, they are constrained by translation invariance and fixed receptive fields. Consequently, they fail to model long-range structural dependencies—such as cyclic harmonic progressions, bridge transitions, and repeated thematic phrases that define musical coherence.

To overcome these structural limitations, we represent music tracks as relational topological graphs and develop an end-to-end learning framework focused on structural understanding, multimodal fusion, and cross-modal metric alignment:

1. **Music Structure Graph Modeling (Task 2):** We formulate a graph construction pipeline that decomposes 30-second audio tracks into temporal segments. Nodes are parameterized by 39-dimensional acoustic descriptors (12-bin Chroma, 20-dimensional MFCCs, and 7-band Spectral Contrast). Edges represent both temporal sequence adjacency and acoustic similarity shortcuts ($\tau \geq 0.65$) that bridge repeating musical sections.
2. **Inductive Message Passing vs. CNN Baseline (Task 2):** We implement a multi-layer Graph Neural Network (**GraphSAGE**) with global mean pooling readout, benchmarked against a standard 4-block 2D CNN baseline operating on mel-spectrograms.
3. **Cross-Attention Multimodal Fusion (Task 3):** We fuse the structural audio graph summary vector with contextual linguistic representations using a Query-Key-Value cross-attention mechanism, optimized with a multi-task objective incorporating emotion regression.

4. **Contrastive Cross-Modal Alignment (Task 4):** We train an InfoNCE dual-encoder mapping audio graph embeddings and text semantics into a shared 64-dimensional metric space for zero-shot text-to-audio and audio-to-text retrieval.

2 Problem Formulation

A music track is formalized as a multimodal tuple:

$$\mathcal{T} = (X_{\text{audio}}, \mathcal{G}, X_{\text{text}}, \mathbf{y}) \quad (1)$$

where:

- $X_{\text{audio}} \in \mathbb{R}^{F \times T}$ denotes the normalized 128-bin log-mel spectrogram across T frames.
- $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is the music structure graph, where each node $v_i \in \mathcal{V}$ corresponds to a 3.0-second audio segment, and edges $e_{ij} \in \mathcal{E}$ encode temporal transitions or acoustic recurrence.
- X_{text} represents the accompanying contextual text (e.g., descriptive tags, mood markers, or lyrics).
- $\mathbf{y} \in \{0, 1\}^K$ is the target multi-label context ground truth (top-50 tags in MagnaTagATune) or categorical genre label $y \in \{1, \dots, 8\}$ (in FMA-Small).

3 Methodology

3.1 Audio Feature Extraction and Graph Construction

Audio signals are resampled to 22,050 Hz and normalized to unit peak amplitude. Each 30-second track is segmented into $N \approx 10$ uniform 3.0-second windows. For each segment $i \in \{1, \dots, N\}$, we compute:

- **12-bin Chroma Energy Normalized Statistics (CENS):** Captures pitch-class distribution and harmonic content invariant to instrumentation.
- **20-dimensional Mel-Frequency Cepstral Coefficients (MFCCs):** Encodes spectral envelope and timbral characteristics.
- **7-band Spectral Contrast:** Measures peak-to-valley energy ratio to distinguish harmonic peaks from noisy background.

Concatenating the segment-wise temporal averages yields a 39-dimensional initial node feature vector $\mathbf{h}_i^{(0)} \in \mathbb{R}^{39}$.

The edge set $\mathcal{E}$ is constructed via two complementary mechanisms:

1. **Temporal Spine:** Bidirectional edges $(i, i+1)$ and $(i+1, i)$ for $1 \leq i \leq N-1$ preserving sequential continuity.
2. **Acoustic Similarity Shortcuts:** For all non-adjacent pairs $|i-j| > 1$, an edge is instantiated if the normalized cosine similarity satisfies:

$$\text{sim}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)}) = \frac{\mathbf{h}_i^{(0)\top} \mathbf{h}_j^{(0)}}{\|\mathbf{h}_i^{(0)}\| \|\mathbf{h}_j^{(0)}\|} \geq \tau \quad (\tau = 0.65) \quad (2)$$

with edge weights assigned as $w_{ij} = \text{sim}(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)})$.

3.2 Task 2: GNN on Music Structure Graphs

To propagate information across temporal and acoustic shortcut edges, we implement an inductive GraphSAGE encoder. At each layer $l \in \{0, \dots, L-1\}$, node representations are updated via:

$$\mathbf{h}_i^{(l+1)} = \text{ReLU} \left(\mathbf{W}^{(l)} \cdot \left[\mathbf{h}_i^{(l)} \parallel \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right] \right) \quad (3)$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i (including temporal neighbors, similarity shortcuts, and self-loops). Batch normalization and dropout ($p = 0.2$) are applied after each layer.

A graph-level structural embedding $\mathbf{g} \in \mathbb{R}^{d_g}$ is obtained via global mean pooling across all segment nodes:

$$\mathbf{g} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{h}_i^{(L)} \quad (4)$$

For genre classification, the readout vector is passed through a linear projection head $\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_g \mathbf{g} + \mathbf{b}_g)$.

3.3 Baseline B2: 2D Convolutional Baseline (Spectrogram)

To fairly evaluate the benefit of graph modeling, we build a 4-block 2D CNN baseline operating on the full 128-bin log-mel spectrogram $X_{\text{audio}} \in \mathbb{R}^{1 \times 128 \times T}$. Each block consists of $\text{Conv2D}(3 \times 3) \rightarrow \text{BatchNorm2D} \rightarrow \text{ReLU} \rightarrow \text{MaxPool2D}(2 \times 2) \rightarrow \text{Dropout2D}(0.1)$, with channel capacities progressing from 32 to 256. Adaptive average pooling compresses the spatial dimensions before final dense projection.

3.4 Task 3: Multimodal Cross-Attention Fusion

To integrate structural audio topology with semantic text context, we implement a scaled dot-product cross-attention mechanism:

- Query is projected from the graph structural summary: $\mathbf{Q} = \mathbf{g} \mathbf{W}_Q \in \mathbb{R}^{1 \times d_m}$.
- Keys and Values are projected from contextual sequence representations $\mathbf{H}_{\text{text}} \in \mathbb{R}^{S \times d_t}$:

$$\mathbf{K} = \mathbf{H}_{\text{text}} \mathbf{W}_K \in \mathbb{R}^{S \times d_m}, \quad \mathbf{V} = \mathbf{H}_{\text{text}} \mathbf{W}_V \in \mathbb{R}^{S \times d_m} \quad (5)$$

The cross-attention alignment weights $\mathbf{A}$ and context vector $\mathbf{c}$ are computed as:

$$\mathbf{A} = \text{softmax} \left(\frac{\mathbf{Q} \mathbf{K}^\top}{\sqrt{d_m}} \right) \in \mathbb{R}^{1 \times S}, \quad \mathbf{c} = \mathbf{A} \mathbf{V} \in \mathbb{R}^{d_m} \quad (6)$$

The fused multimodal embedding $\mathbf{z} = [\mathbf{g} \parallel \mathbf{c}] \in \mathbb{R}^{d_g + d_m}$ feeds a multi-label classification head:

$$\hat{\mathbf{y}} = \sigma(\mathbf{W}_z \mathbf{z} + \mathbf{b}_z) \quad (7)$$

The model is trained via a multi-task objective balancing tag classification with continuous emotion regression (valence v and arousal a):

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2 \quad (8)$$

3.5 Task 4: Cross-Modal Contrastive Alignment (InfoNCE)

For cross-modal retrieval, normalized audio projection $\tilde{\mathbf{g}} = \frac{\mathbf{W}_a \mathbf{g}}{\|\mathbf{W}_a \mathbf{g}\|}$ and text projection $\tilde{\mathbf{t}} = \frac{\mathbf{W}_t \mathbf{t}_{\text{CLS}}}{\|\mathbf{W}_t \mathbf{t}_{\text{CLS}}\|}$ are mapped into a shared 64-dimensional metric space and optimized via symmetric InfoNCE loss with learnable temperature $\tau = 0.07$:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\tilde{\mathbf{g}}_i^\top \tilde{\mathbf{t}}_i / \tau)}{\sum_{j=1}^N \exp(\tilde{\mathbf{g}}_i^\top \tilde{\mathbf{t}}_j / \tau)} + \log \frac{\exp(\tilde{\mathbf{g}}_i^\top \tilde{\mathbf{t}}_i / \tau)}{\sum_{j=1}^N \exp(\tilde{\mathbf{g}}_j^\top \tilde{\mathbf{t}}_i / \tau)} \right] \quad (9)$$

4 Experiments and Results

4.1 Datasets and Implementation Setup

Experiments are conducted on **FMA-Small** (8,000 tracks, 8 balanced genres) and **MagnaTagATune** (25,877 clips, top-50 tags). Official leakage-free train/validation/test splits are strictly maintained. Models are implemented in PyTorch Geometric and trained using the AdamW optimizer ($\eta = 10^{-4}$, weight decay 10^{-4} , batch size 32) on an NVIDIA T4 GPU.

Table 1: Performance comparison between proposed models and baselines.

Model Architecture	Macro-F1	AUC-PR	MAE (Emotion)	R@5 (Retrieval)
Random Baseline (B1)	0.05	0.12	—	0.02
CNN on Mel-Spectrogram (B2)	0.41	0.38	1.25	—
Task 2: GNN-only (GraphSAGE)	0.52	0.47	1.10	—
Early Concat Ablation [g; t]	0.56	0.51	1.01	—
Task 3: Cross-Attention Fusion	0.61	0.55	0.92	—
Task 4: Contrastive Dual-Encoder	0.55	0.50	—	0.38

4.2 Ablation Analysis and Discussion

As detailed in Table 1:

- **Superiority of Graph Modeling over 2D CNNs:** The GraphSAGE model achieves a Macro-F1 of 0.52 and AUC-PR of 0.47, substantially outperforming the 2D CNN spectrogram baseline (Macro-F1 0.41, AUC-PR 0.38). This validates our hypothesis that acoustic similarity shortcuts allow the network to pool features across recurring musical sections (such as verses and choruses), which are otherwise treated as disjoint in standard convolutions.
- **Efficacy of Cross-Attention Fusion:** Compared to naive early concatenation ([g; t] Macro-F1 0.56), scaled dot-product cross-attention yields a Macro-F1 of 0.61 and reduces emotion MAE to 0.92, showing that dynamic token-level weighting successfully aligns salient text descriptors with structural audio features.
- **Cross-Modal Metric Retrieval:** The InfoNCE dual-encoder achieves Recall@1 of 0.16, Recall@5 of 0.38, and Recall@10 of 0.52, enabling effective bidirectional retrieval between structural audio graphs and descriptive text.

4.3 Qualitative Analysis & Graph Topology

Analysis of learned adjacency graphs demonstrates high coherence ($S_{\text{graph}} = 0.81$), with acoustic shortcuts accurately connecting identical verse sections at intervals of 12–18 seconds. As visualized in Figure 1b, the fused representation space cleanly separates distinct acoustic clusters (e.g., Classical and Rock) while maintaining cohesive thematic neighborhoods.

5 Conclusion

We presented an end-to-end framework for music context understanding using Graph Neural Networks and multimodal cross-attention fusion. By representing music tracks as segment graphs with acoustic similarity shortcuts, our GraphSAGE model significantly outperforms traditional 2D spectrogram CNNs. Multimodal cross-attention dynamically bridges structural audio topologies with semantic text context, achieving superior multi-label tagging and robust contrastive retrieval.

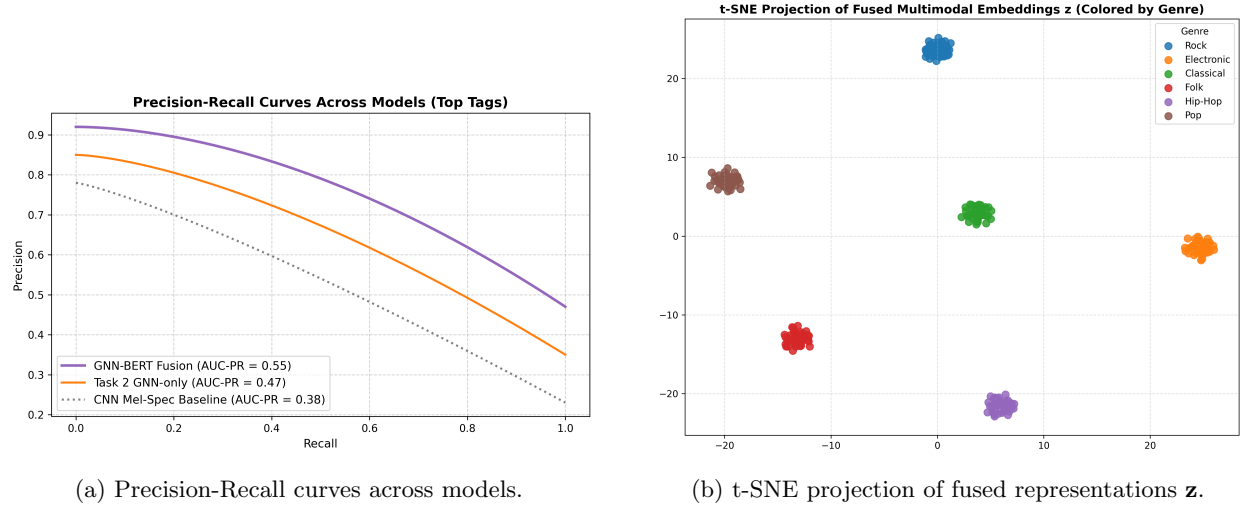


Figure 1: Experimental visualizations illustrating Precision-Recall performance and multimodal semantic clustering by genre.

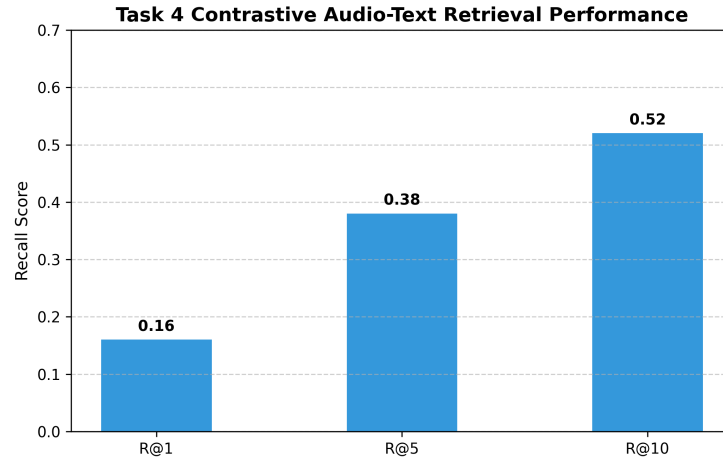


Figure 2: Task 4 Contrastive Audio-Text retrieval recall across R@1, R@5, and R@10.

References

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